

Question:

Four days after undergoing a Whipple procedure for newly-diagnosed pancreatic cancer, a 65-year-old man has shortness of breath. His surgery was complicated by bleeding for which he required intraoperative transfusion with 4 units of packed red blood cells and 1 unit of platelets. His temperature is 38.8°C (101.8°F), pulse is 110/min, respirations are 26/min, and blood pressure is 95/55 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 85%. Cardiac examination shows normal heart sounds and no jugular venous distention. Auscultation of the lungs shows diffuse crackles bilaterally. The extremities are warm and there is no edema. Laboratory studies show a leukocyte count of 17,000/mm³ and hemoglobin concentration of 9.8 g/dL. Arterial blood gas on room air shows:pH 7.35PaO₂ 41 mm HgPaCO₂ 38 mm HgHCO₃⁻ 25 mEq/LThe patient is intubated and mechanical ventilation is initiated. An x-ray of the chest is shown. Transthoracic echocardiography shows a normally contracting left ventricle. Which of the following is the most likely cause of this patient's current condition?"Answer Choices:A. Decreased chest wall complianceB. Formation of anti-leukocyte antibodiesC. Diffuse inflammatory alveolar damageD. Increased left atrial pressures

Reasoning + Answer:

<think>the patient has the fever, dyspnea, tachycardia, hypotension, elevated co2. the arterial blood gas shows a high paCO2 which suggests respiratory acidosis. a transthoracic echocardiogram shows a normally contracting left ventricle but pulmonary crackles. those are both indicating there is alveolar edema. dyspnea and shortness of breath is also hint for this. the 4 units of packed red blood cells and 1 unit of platelets suggests bleeding. all this information suggests that patient has ALI (acute lung injury) </think> <answer> C. Diffuse inflammatory alveolar damage </answer>

